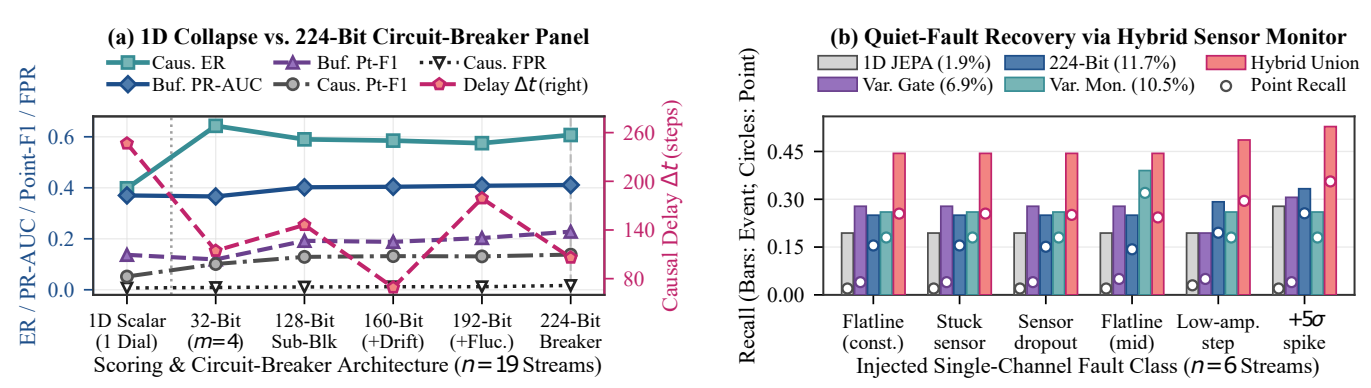

(a) 1D Collapse vs. 224-Bit Circuit-Breaker Panel
Caus. ER
Buf. PR-AUC
Buf. Pt-F1
Caus. Pt-F1
Caus. FPR
Delay Δt (right)
ER / PR-AUC / Point-F1 / FPR
Causal Delay Δt (steps)
0.0
0.2
0.4
0.6
80
140
200
260
1D Scalar (1 Dial)
32-Bit (m=4)
128-Bit Sub-Blk
160-Bit (+Drift)
192-Bit (+Fluc.)
224-Bit Breaker
Scoring & Circuit-Breaker Architecture (n = 19 Streams)
(b) Quiet-Fault Recovery via Hybrid Sensor Monitor
1D JEPA (1.9%)
224-Bit (11.7%)
Hybrid Union
Var. Gate (6.9%)
Var. Mon. (10.5%)
Point Recall
Recall (Bars: Event; Circles: Point)
0.00
0.15
0.30
0.45
Flatline (const.)
Stuck sensor
Sensor dropout
Flatline (mid)
Low-amp. step
+5σ spike
Injected Single-Channel Fault Class (n = 6 Streams)